

Topic 07: Homogeneous First-Order Equations & Geometric Substitutions

Professor Leonard Master Series · Lessons 20 & 21

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Scaling Symmetry

"What does 'homogeneous' mean here? It means that every term in your equation has the EXACT same total degree of variables!"

For example, look at $(x^2 + y^2)dx - 2xydy = 0$:

- x^2 has degree 2.
- y^2 has degree 2.
- $2xy$ has degree $1 + 1 = 2$.

Because all terms scale equally under zoom ($x \rightarrow tx, y \rightarrow ty$), the slope $\frac{dy}{dx}$ depends **only on the ratio** $\frac{y}{x}$! If we set $u = \frac{y}{x}$, the equation collapses into a simple separable equation!

2. The Substitution Machine: $y = ux$

Whenever you recognize a homogeneous first-order equation:

1. Let $y = u(x) \cdot x$.
2. By the Product Rule:

$$\frac{dy}{dx} = u + x \frac{du}{dx} \quad (\text{or } dy = udx + xdu)$$

3. Substitute both y and $\frac{dy}{dx}$ into your equation.
4. All the x 's will factor out and cancel, leaving a clean separable equation for u and x !
5. Solve for u , then remember to back-substitute $u = \frac{y}{x}$.